

Course 1 · Week 1 — Toolchain & data hygiene

Cheatsheet — biostats_courses

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The research workflow

Question → Measurements → Design → Acquisition → Description → Analysis → Interpretation → Validation → Knowledge

Biological vs statistical significance: “ $p < 0.05$ ” ≠ “effect matters”.

R / RStudio / Quarto / renv

Task	Command
Install packages	<code>install.packages("pkg")</code>
Pin project	<code>renv::init() / renv::snapshot() / renv::restore()</code>
Render one file	<code>quarto render path/to/file.qmd</code>
Preview with live reload	<code>quarto preview</code>
Render only slides	<code>quarto render file.qmd --to revealjs</code>

Data types and scales

Scale	Example	Statistic
Nominal	sex, site	counts, proportions
Ordinal	pain 0–10	median, IQR, rank tests
Interval	°C	mean, SD
Ratio	kg, time	mean, SD, ratios

Accuracy ≠ precision. Bias shifts the target; variance spreads the shots.

Tidy data

1. One variable per column.
2. One observation per row.
3. One observational unit per table.

dplyr verbs

```
df |>
  filter(age >= 18) |>
  select(id, age, sbp) |>
  mutate(sbp_z = (sbp - mean(sbp)) / sd(sbp)) |>
  group_by(arm) |>
  summarise(mean_sbp = mean(sbp, na.rm = TRUE), n = n())
```

Joins: `left_join`, `inner_join`, `anti_join` by a key.

Missingness: `is.na()`, `drop_na()`, `tidyr::complete()`.

ggplot2 grammar

```
ggplot(data, aes(x, y, colour = group)) +  
  geom_point() +  
  facet_wrap(~ factor) +  
  labs(x = "x label", y = "y label") +  
  theme_minimal()
```

Layer order: data → aes → geom → facet → scales → labels → theme.

Plot families

- One variable continuous → **histogram**, **density**.
- Two continuous → **scatter**, sometimes binned / smoothed.
- One categorical + one continuous → **boxplot**, **violin**, **strip**.
- Two categorical → **bar with fill**, **mosaic**.

Decision rule for Week 1

- Cannot read the data into a tibble → fix data entry first.
- Tibble reads but has missing or mixed types → clean before modelling.
- Plot is ugly or cluttered → trust the plot, not the summary statistic.

Common pitfalls

- Running statistics on untyped / partially-missing data.
- Adjusting colour scales instead of simplifying the chart.
- Reporting mean + SD for a skewed variable.
- Forgetting `set.seed()` in any simulation chunk.

Further reading

- Wickham, *R for Data Science* (2e).
- Posit ggplot2 and dplyr cheatsheets.